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ORIGINAL RESEARCH ARTICLE

Thermodynamic Modeling of the Te-*X* (*X* = Gd, Dy, Ho) Binary Systems Combined with the First-Principles Method

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Abstract Thermodynamic databases are essential to understanding alloy properties and guiding materials design. In this work, the mixing enthalpies of the liquid phase in the Gd-Te, Dy-Te and Ho-Te binary systems are calculated using Ab-initio Molecular Dynamics (AIMD) simulations and the thermodynamic parameters are determined using the CALculation of PHAse Diagrams (CALPHAD) method combined with the phase equilibrium data and thermodynamic data. The associated solution model is employed to describe the liquid phase of these systems. The sublattice model $(Gd,Te)_{0.0296}(Gd)_{0.4}(Te)_{0.5714}$, $(Dy,Te)_{0.0286}(Dy)_{0.4286}(Te)_{0.5714}$ and $(Ho,Te)_{0.5}(Te)_{0.5}$ are utilized to describe Gd_2Te_3, Dy_2Te_3 and HoTe, respectively. The line compounds, i.e. GdTe, DyTe, and $DyTe_2$, are modeled using the stoichiometric model. The calculated results can describe the experimental and thermodynamic information reported in the literature. In addition, the existence of a liquid-liquid separation in the Dy-rich side in the Dy-Te binary system is proposed in this work.

Keywords AIMD · CALPHAD · Dy-Te · Gd-Te · Ho-Te

1 Introduction

Phase-Change Memory (PCM) materials store information through the large contrast in electrical resistance of phase change materials between the amorphous and crystalline state, which is considered as the next generation in non-volatile electronic data storage because of the advantage of SET (crystallization/write) speed, RESET (amorphization/erase) speed and thermodynamic stability over traditional memory materials.[1–5] Ge-Te, Sb-Te and Ge-Sb-Te materials are extensively researched as PCM phase-change materials. The thermodynamic databases and outputs of the theoretical calculations are important for multi-scale simulations, which can be used as the input information of phase field simulations, diffusion processes modelling, phase transformations, material properties and structure morphology development.[6] To study the principles of doping elements in phase change materials and guide the materials design of PCM, a part of the thermodynamic databases of Te-base and Sb-base materials is assessed by our group.[7–10]

Many researchers report that the storage property of phase-change materials can be increased by doping with rare-earth (RE) elements.[11–13] For example, the thermal stability is improved by doping with Gd.[13] For the development of Ge-Sb-Te phase-change materials as well as for the materials design of the RE-doped phase-change materials, further study of the thermodynamic description of the phases in the RE-Te systems is required to be carried out basing on the current results of our group. Currently, the thermodynamic description of RE-Te systems is not fully studied. Therefore, building thermodynamic databases of RE-Te systems is necessary.

The aim of this work is to extend the thermodynamic descriptions of the Gd-Te, Dy-Te and Ho-Te binary systems and support the calculation of a wider range of alloys in the complete ternary system in the future. A critical evaluation of all available experimental data in the literature was performed in conjunction with the first-principles method and the CALculation of PHAse Diagrams

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(CALPHAD)[14] method modeling to generate complete binary thermodynamic descriptions.

2 Literature Review

The published research of the Gd-Te, Dy-Te and Ho-Te was reviewed. The phase equilibrium information of all these binary systems and most of the thermodynamic properties are reported in the literature. The crystal information of the phases in these binary systems is summarized in Table 1 according to the literature.[15–30]

2.1 The Gd-Te Binary System

The Gd-Te binary phase diagram was measured using X-ray diffraction (XRD) and thermal analysis methods by Abrikosov and Zargarian.[31] There are 6 compounds in the Gd-Te bianry system and no compound is found in the range of 0–50 at.% Te. GdTe has the highest melting point and congruently melts at 2098 K. The homogeneity range of Gd_2Te_3 is also reported[31] and it ranges from 57.14 to 60 at.% Te. Herle and Corbett[26] report that the Gd_2Te exists stably below 1323 K and decomposes into GdTe and Gd above this temperature. However, experimental details are not provided. Therefore, the reaction, hcp(Gd) + GdTe ↔ Gd_2Te, is not considered in this work.

In addition, the formation enthalpy and formation entropy of GdTe, Gd_4Te_7, $GdTe_2$, Gd_2Te_5 and $GdTe_3$ are determined using electromotive force (EMF) method by Azizov et al.[32] The formation enthalpies of GdTe, Gd_2Te_3 and $GdTe_2$ are calculated using the first-principles method by Jain et. al.[33] and GdTe is calculated using Miedema model by Eichler et al.[34] However, the mixing enthalpy of the Gd-Te system is not reported at present.

2.2 The Dy-Te Binary System

The Dy-Te phase diagram was measured by Abrikosov and Zinchenko,[27] in which 6 compounds, Dy_2Te, DyTe, Dy_2Te_3, Dy_4Te_7, Dy_4Te_9 and Dy_4Te_{11}, exist. The crystal structures of these compounds were also determined by

Table 1 The crystal structure of terminal components and intermediate compounds in Te-*X* (*X* = Gd, Ho or Te) systems[15–30]

Phase	Space group	Prototype	Pearson symbol	Lattice parameter, nm	Ref
Pure elements					
bcc(Gd)	$Im\bar{3}m$	W	*cI*2	*a* = 0.405	[15]
hcp(Gd)	$P6_3/mmc$	Mg	*hP*2	*a* = 0.636, *c* = 0.578	[16]
hcp(Dy)	$Im\bar{3}m$	Mg	*cI*2	*a* = 0.3980	[17]
bcc(Dy)	$P6_3/mmc$	W	*hP*2	*a* = 3.5930 *c* = 5.6537	[18]
hcp(Ho)	$P6_3/mmc$	Mg	*hR*2	*a* = 0.358, *c* = 0.562	[19]
hex(Te)	*P*3121	γSe	*hP*3	*a* = 0.445, *c* = 0.593	[20]
Gd-Te binary system					
GdTe	$Fm\bar{3}m$	NaCl	*cF*8	*a* = 0.432, *c* = 0.89	[21]
Gd_2Te_3	*Pnma*	Sb_2S_3	*oP*20	*a* = 1.201, *b* = 0.4301, *c* = 1.182	[22]
Gd_4Te_7	*P*4/*nmm*	Cu_2Sb	*tP*6	*a* = 0.612	[23]
$GdTe_2$	*P*4/*nmmZ*	Fe_2As	*tP*6	*a* = 0.432, *c* = 0.895	[24]
Gd_2Te_5	*Cmcm*	Nd_2Te_5	*oC*28	*a* = 0.434, *b* = 4.36, *c* = 0.434	[25]
$GdTe_3$	*Cmcm*	$NdTe_3$	*oC*16	*a* = 0.432, *b* = 2.528 *c* = 0.433	[25]
Dy-Te binary system					
Dy_2Te	*Pnma*	Sc_2Te	*oP*36	*a* = 2.1922, *b* = 0.465 *c* = 1.1428	[26]
DyTe	*Fm*3*m*	NaCl	*cF*8	*a* = 0.6077	[27]
Dy_2Te_3	*Fddd*	Sc_2S_3	*oF*80	*a* = 1.220, *b* = 0.8645, *c* = 2.585	[27]
Dy_4Te_7	*P*4/*mbm*	Cu_2Sb	*tP*6	*a* = 0.4274, *c* = 0.8917	[27]
Dy_4Te_9	*Bmmb*	Nd_2Te_5	-	*a* = 0.4304, *b* = 0.4301, *c* = 4.320	[27]
Dy_4Te_{11}	*Bmmb*	$NdTe_3$	-	*a* = 0.4298, *b* = 0.4298, *c* = 2.542	[27]
Ho-Te binary system					
HoTe	$Fm\bar{3}m$	NaCl	*cF*8	*a* = 0.607	[28]
Ho_2Te_5	*Cmcm*	Nd_2Te_5	*oC*28	*a* = 0.441 *b* = 4.503 *c* = 0.441	[29]
$HoTe_3$	*Cmcm*	$NdTe_3$	*oC*16	*a* = 0.429 *b* = 2.540 c = 0.429	[30]

Abrikosov and Zinchenko.[27] According to the measured phase diagram, there are two congruent melting compounds in the Dy-Te binary system: DyTe and Dy_2Te_3, which melt at 2123 and 1846 K, respectively. Later, Herle and Corbett[26] experimentally demonstrated the existence of Dy_2Te and reported that Dy_2Te decomposes into DyTe and hcp(Dy) above about 1343 K. Herle and Corbett[26] also determined that Dy_2Te has the space group *Pnma*.

There are a few reports on the thermodynamic data of the Dy-Te binary system in the literature. Eichler et al.[34] and Jain et al.[33] calculated formation enthalpies of a part of the compounds in the Dy-Te binary system. No mixing enthalpy data has been reported.

2.3 The Ho-Te Binary System

Yarembash et al.[29] measured the Ho-Te binary phase diagram and found 3 compounds in this binary system, which are HoTe, Ho_2Te_5 and $HoTe_3$. HoTe has a homogeneity range and melts congruently at about 1643 K. Very little has been reported on the thermodynamic data in the literature. The formation enthalpy of HoTe was also calculated using Miedema model by Eichler et al.[34] Later, Ghamri et al.[35] calculated the formation enthalpies of HoTe and Ho_2Te_5, which is close to the value reported by Eichler et al.[34] The Ho-Te binary phase diagram was assessed using CALPHAD method by Ghamri et al.[35] without thermodynamic data on the mixing enthalpies. However, the mixing enthalpies calculated using the thermodynamic parameters from Ghamri et al.[35] are different from the results calculated using ab-initio Molecular Dynamics (AIMD) simulations in this work.

3 Calculation Details

In this work, the AIMD simulations are employed to calculate the mixing enthalpies of the Te-*X* binary system and the density functional theory (DFT) method is used to calculate the formation enthalpies of the compounds in the Dy-Te binary system. In addition, the thermodynamic parameters of these binary systems are determined using the CALPHAD method.

3.1 AIMD Assumulations

The AIMD simulations for molten $X_{1\text{-}m}Te_m$ (where X = Gd, Dy and Ho, and m is the mole fraction of Te) are carried out using Vienna Ab-initio Simulation Package (VASP)[36–38] with the exchange-correlation functional described by the generalized gradient approximation (GGA) modified by Perdew, Burke and Ernzerhof[39] (PBE). Cubic supercells containing 216 atoms with periodic boundary and the canonical (or NVT) ensemble (fixing the number of atoms, volume, and temperature), are employed for these AIMD simulations. The simulation temperature for each supercell is 2280 K, which is above the liquidus temperatures in the Te-*X* systems. The plane-wave cutoff energy is set as 300 eV. The Brillouin zone is sampled at a single gamma-point. All the AIMD simulations are run for 16,000 steps with a time step of 3 femtoseconds. The first 6,000 steps are used to reach kinetic equilibrium in the AIMD simulations, and the configurations from the remaining 10,000 steps are employed to analyze the thermodynamic properties. In the present AIMD simulations, the volumes of the supercell sizes as a function of composition are adjusted carefully to ensure that the total external pressure is close to zero.

The mixing enthalpies of molten $X_{1\text{-}m}Te_m$ at 2280 K are calculated by the following formula:

$$\Delta E_{X_{1-m}Te_m} = E_{X_{1-m}Te_m} - mE_{Te} - (1-m)E_X \quad \text{(Eq 1)}$$

where $E_{X_{1-m}\mathrm{Te}_m}$ is the total energy of system at the different compositions, m is the mole fraction of component Te, E_{Te} the enthalpy of pure component Te and E_X the enthalpy of pure component X (X = Gd, Dy and Ho).

3.2 DFT Calculations

Because the compound Dy_2Te is considered in this work and the formation enthalpy of Dy_2Te is not reported in the literature, the enthalpies of formation of Dy_2Te, DyTe and Dy_2Te_3 are calculated using DFT[40] with VASP.[36–38] Because the formation enthalpies of compounds in the Gd-Te and Ho-Te binary systems are reported in the literature, they are not calculated in this work. The PBE generalized gradient approximation[39] is chosen as the exchange and correlation pseudopotential function. The cutoff energy for all elements and compounds can be selected as 450 eV after convergence tests. The *k*-points files are generated according to the Monkhorst-Pack scheme.[41] The magnetism testing of pure elements are done in this work. Different magnetic moments are given to calculate the total energy and the total energy did practically not change. According to the total energy obtained at the beginning, the enthalpy of formation of the compound Dy_aTe_b is calculated as follows:

$$\Delta_f E(Dy_aTe_b) = E(Dy_aTe_b) - \frac{a}{a+b}E(Dy) - \frac{b}{a+b}E(Te) \quad \text{(Eq 2)}$$

in which $E(\mathrm{Dy}_a\mathrm{Te}_b)$, $E(\mathrm{Dy})$ and $E(\mathrm{Te})$ are equilibrium energies of the intermetallic compound $\mathrm{Dy}_a\mathrm{Te}_b$ and the pure elements of Dy and Te, respectively.

3.3 Thermodynamic Models

The Gibbs energy function ${}^0G_i(T)$ for the pure elements i, i.e. Gd, Dy, Ho and Te, are expressed as:

$$ {}^0G_i = a + bT + cT\ln T + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9} \quad \text{(Eq 3)} $$

where T is the absolute temperature and a to h are coefficients. In present modeling, the values of a to h are taken from the Scientific Group Thermodata Europe (SGTE) compilation by Dinsdale.[42]

The Gibbs energy of the liquid phase is described using the associate model because the value of mixing enthalpy decreases rapidly with composition and the liquidus is not symmetrical on both sides of the compounds XTe (X = Gd, Dy or Ho). The associate model, $(X, X_{0.5}\mathrm{Te}_{0.5}, \mathrm{Te})_1$, is used to describe the liquid phase in the X-Te system (X = Gd, Dy and Te). The corresponding stoichiometry, $X_{0.5}\mathrm{Te}_{0.5,}$ is treated as an associate in this work. The Gibbs energy of the liquid phase is:

$$ \begin{aligned} G^{\mathrm{Liquid}} = & y_X^0 G_X^{\mathrm{Liquid}} + y_{\mathrm{Te}}^0 G_{\mathrm{Te}}^{\mathrm{Liquid}} + y_{X\mathrm{Te}}^0 G_{X\mathrm{Te}}^{\mathrm{Liquid}} \\ & + RT[y_X ln(x_X) + y_{\mathrm{Te}} ln(x_{\mathrm{Te}}) + y_{X\mathrm{Te}} ln(y_{X\mathrm{Te}})] \\ & + y_X y_{X\mathrm{Te}}({}^0L_{X,X\mathrm{Te}}^{\mathrm{Liquid}} + {}^1L_{X,X\mathrm{Te}}^{\mathrm{Liquid}}(y_X - y_{X\mathrm{Te}}) + {}^2L_{X,\mathrm{Te}}^{\mathrm{Liquid}}(y_X - y_{X\mathrm{Te}})^2) \\ & + y_{X\mathrm{Te}} y_{\mathrm{Te}}\left({}^0L_{X\mathrm{Te},\mathrm{Te}}^{\mathrm{Liquid}} + {}^1L_{X\mathrm{Te},\mathrm{Te}}^{\mathrm{Liquid}}(y_{X\mathrm{Te}} - y_{\mathrm{Te}}) + {}^2L_{X\mathrm{Te},\mathrm{Te}}^{\mathrm{Liquid}}(y_{X\mathrm{Te}} - y_{\mathrm{Te}})^2\right) \end{aligned} \quad \text{(Eq 4)} $$

where R is the gas content. ${}^0G_X^{\mathrm{liquid}}$, ${}^0G_{\mathrm{Te}}^{\mathrm{liquid}}$ and ${}^0G_{X\mathrm{Te}}^{\mathrm{liquid}}$ are the Gibbs energies for pure X (X = Gd, Dy or Ho), Te and corresponding associate at the liquid state. The parameter ${}^0G_X^{\mathrm{liquid}}$ and ${}^0G_{\mathrm{Te}}^{\mathrm{liquid}}$ are also taken from the SGTE database[42] and ${}^0G_{X\mathrm{Te}}^{\mathrm{liquid}}$ is determined in this work.y_X, y_{Te} and $y_{X\mathrm{Te}}$ are the mole fractions for X, Te and XTe, respectively. ${}^vL_{i,j}^{\mathrm{liquid}}$($i$, j = X, Te and XTe) are the interaction parameters from binary system. In addition, interaction parameters can be described by:

$$ L = a + bT \quad \text{(Eq 5)} $$

where a and b are contents to be evaluated.

The sublattice model, $(\mathrm{Gd,Te})_{0.0296}(\mathrm{Gd})_{0.4}(\mathrm{Te})_{0.5714}$, $(\mathrm{Dy,Te})_{0.0286}$ $(\mathrm{Dy})_{0.4286}(\mathrm{Te})_{0.5714}$ and $(\mathrm{Ho,Te})_{0.5}(\mathrm{Te})_{0.5}$ are used to describe Gd_2Te_3, Dy_2Te_3 and HoTe, respectively, because these binary compounds have a homogeneity range. Taking HoTe as an example, the Gibbs energy of HoTe can be expressed as:

$$ \begin{aligned} G^{\mathrm{HoTe}} = & y'_{\mathrm{Ho}} G_{\mathrm{Ho:Te}}^{\mathrm{HoTe}} + y'_{\mathrm{Te}} G_{\mathrm{Te:Te}}^{\mathrm{HoTe}} + mRT(y'_{\mathrm{Ho}} \ln y'_{\mathrm{Ho}} + y'_{\mathrm{Te}} \ln y'_{\mathrm{Te}}) \\ & + y'_{\mathrm{Ho}} y'_{\mathrm{Te}}({}^0L_{\mathrm{Ho,Te:Te}}^{\mathrm{HoTe}} + {}^1L_{\mathrm{Ho,Te:Te}}^{\mathrm{HoTe}}(y'_{\mathrm{Ho}} - y'_{\mathrm{Te}}) + {}^2L_{\mathrm{Ho,Te:Te}}^{\mathrm{HoTe}}(y'_{\mathrm{Ho}} - y'_{\mathrm{Te}})^2) \end{aligned} \quad \text{(Eq 6)} $$

where $y_{\mathrm{Ho}}\prime$ and $y_{\mathrm{Te}}\prime$ are the site fractions of X and Te in the first sub-lattice, respectively. m is the ratio of the first sub-lattice, which contains two components.$G_{\mathrm{Ho:Te}}^{\mathrm{HoTe}}$ and $G_{\mathrm{Te:Te}}^{\mathrm{HoTe}}$ are the Gibbs energy of the end-member components. ${}^0L_{\mathrm{Ho,Te:Te}}^{\mathrm{HoTe}}$, ${}^1L_{\mathrm{Ho,Te:Te}}^{\mathrm{HoTe}}$,${}^2L_{\mathrm{Ho,Te:Te}}^{\mathrm{HoTe}}$ are the interaction parameters of HoTe, which also can be described by Eq 5.

For line compounds in the Te-X system, for example, GdTe, DyTe, and $DyTe_2$, the stoichiometric model, $(X)_m(\mathrm{Te})_n$, is used. The Gibbs energy of each line compounds per mole atoms can be expressed as:

$$ G^{X_m\mathrm{Te}_n} = x_X^0 G_X^{\mathrm{SER}} + x_{\mathrm{Te}}^0 G_{\mathrm{Te}}^{\mathrm{SER}} + a + bT \quad \text{(Eq 7)} $$

where x_X and x_{Te} are the mole fractions of components X and Te, respectively. ${}^0G_X^{\mathrm{SER}}$ and ${}^0G_{\mathrm{Te}}^{\mathrm{SER}}$ are the Gibbs energies for X and Te at their standard element reference state respectively. a and b are unknown parameters which are to be evaluated in this work.

4 Results and Discussion

In the present work, the thermodynamic parameters of the Gd-Te, Dy-Te and Ho-Te binary systems are determined using CALPHAD method based on the phase equilibrium information and thermodynamic data, including the AIMD results obtained in this work. The thermodynamic parameters are determined using the PARROT module of the Thermo-Calc software.[43] These assessed parameters are listed in Table 2.

4.1 The Gd-Te Binary System

Figure 1 shows the calculated Gd-Te phase diagram using the assessed thermodynamic parameters from the present work compared with the experimental data measured by Abrikosov and Zargarian.[31] Meanwhile, the calculated equilibrium temperatures and compositions of invariant reactions are list in Table 3. The calculated phase diagram agrees reasonably with the measured data. The calculated temperature of invariant reactions has only a maximum margin of error of 1%. The asymmetrical liquidus and the homogeneity range of Gd_2Te_3 has also been reproduced in this work.

Table 2 Summary of the thermodynamic parameters of the Gd-Te, Dy-Te and Ho-Te systems

Phase	Parameters	Ref
Gd-Te binary system		
Liquid	$(Gd, Gd_{0.5}Te_{0.5}\ Te)_1$	
	${}^0G^{Liquid}_{GdTe} = +0.5G^{liq}_{Gd} + 0.5G^{liq}_{Te} - 125457 + 18.059T$	This work
	${}^0G^{Liquid}_{Gd,GdTe} = +52991 - 13.613T$	This work
	${}^1G^{Liquid}_{Gd,GdTe} = +36155 - 20.572T$	This work
	${}^2G^{Liquid}_{Gd,GdTe} = +5631$	This work
	${}^0G^{Liquid}_{GdTe,Te} = -24375 + 9.049T$	This work
	${}^1G^{Liquid}_{GdTe,Te} = -55 - 5.056T$	This work
	${}^2G^{Liquid}_{GdTe,Te} = -12339 - 3.304T$	This work
GdTe	$(Gd)_{0.5}(Te)_{0.5}$	
	${}^0G^{GdTe}_{Gd:Te} = +0.5G^{SER}_{Gd} + 0.5G^{SER}_{Te} - 151700 + 20.881T$	This work
Gd_2Te_3	$(Gd)_{0.4}(Te)_{0.5714}(Gd,Te)_{0.0296}$	
	${}^0G^{Gd_2Te_3}_{Gd:Te:Gd} = +0.4296G^{SER}_{Gd} + 0.5714G^{SER}_{Te} - 139339 + 20.918T$	This work
	${}^0G^{Gd_2Te_3}_{Gd:Te:Te} = +0.4G^{SER}_{Gd} + 0.6G^{SER}_{Te} - 134080 + 20.930T$	This work
Gd_4Te_7	$(Gd)_{0.3636}(Te)_{0.6364}$	
	${}^0G^{Gd_4Te_7}_{Gd:Te} = +0.3636G^{SER}_{Gd} + 0.6364G^{SER}_{Te} - 124650 + 19.200T$	This work
$GdTe_2$	$(Gd)_{0.3333}(Te)_{0.6667}$	
	${}^0G^{GdTe_2}_{Gd:Te} = +0.3333G^{SER}_{Gd} + 0.6667G^{SER}_{Te} - 116758 + 18.373T$	This work
Gd_2Te_5	$(Gd)_{0.2857}(Te)_{0.7143}$	
	${}^0G^{Gd_2Te_5}_{Gd:Te} = +0.2857G^{SER}_{Gd} + 0.7143G^{SER}_{Te} - 101341 + 15.012T$	This work
$GdTe_3$	$(Gd)_{0.25}(Te)_{0.75}$	
	${}^0G^{GdTe_3}_{Gd:Te} = +0.25G^{SER}_{Gd} + 0.75G^{SER}_{Te} - 89810 + 12.835T$	This work
Dy-Te binary system		
Liquid	$(Dy,Dy_{0.5}Te_{0.5},Te)_1$	
	${}^0G^{Liquid}_{DyTe} = +0.5G^{liq}_{Dy} + 0.5G^{liq}_{Te} - 142520 + 31.22T$	This work
	${}^0G^{Liquid}_{Dy,DyTe} = +122926 - 49.883T$	This work
	${}^1G^{Liquid}_{Dy,DyTe} = -18047 + 10.065T$	This work
	${}^2G^{Liquid}_{Dy,DyTe} = +75154 - 36.061T$	This work
	${}^0G^{Liquid}_{DyTe,Te} = -28100 - 19.872T$	This work
	${}^1G^{Liquid}_{DyTe,Te} = -146260 + 79.273T$	This work
	${}^2G^{Liquid}_{DyTe,Te} = -18646 - 27.990T$	This work
Dy_2Te	$(Dy)_{0.6667}(Te)_{0.3333}$	
	${}^0G^{Dy_2Te}_{Dy:Te} = +0.6667G^{SER}_{Dy} + 0.3333G^{SER}_{Te} - 98800 + 15.981T$	This work
DyTe	$(Dy)_{0.5}(Te)_{0.5}$	
	${}^0G^{DyTe}_{Dy:Te} = +0.5G^{SER}_{Dy} + 0.5G^{SER}_{Te} - 144253 + 20.800T$	This work
Dy_2Te_3	$(Dy)_{0.4286}(Te)_{0.5714}(Dy,Te)_{0.0286}$	
	${}^0G^{Dy_2Te_3}_{Dy:Te:Dy} = +0.4286G^{SER}_{Dy} + 0.5714G^{SER}_{Te} - 147982 + 21.296T$	This work
	${}^0G^{Dy_2Te_3}_{Dy:Te:Te} = +0.4G^{SER}_{Dy} + 0.6G^{SER}_{Te} - 137923 + 17.504T$	This work
Dy_4Te_7	$(Dy)_{0.3636}(Te)_{0.6364}$	
	${}^0G^{Dy_4Te_7}_{Dy:Te} = +0.3636G^{SER}_{Dy} + 0.6364G^{SER}_{Te} - 124865 + 12.647T$	This work
Dy_4Te_9	$(Dy)_{0.3077}(Te)_{0.6923}$	
	${}^0G^{Dy_4Te_9}_{Dy:Te} = +0.3077G^{SER}_{Dy} + 0.6923G^{SER}_{Te} - 105006 + 7.882T$	This work

Table 2 continued

Phase	Parameters	Ref
Dy_4Te_{11}	$(Dy)_{0.2667}(Te)_{0.7333}$	
	${}^0G^{Dy_4Te_{11}}_{Dy:Te} = +0.2667G^{SER}_{Dy} + 0.7333G^{SER}_{Te} - 90560 + 4.788T$	This work
Ho-Te binary system		
Liquid	$(Ho, Ho_{0.5}Te_{0.5}, Te)_1$	
	${}^0G^{Liquid}_{HoTe} = +G^{liq}_{Te} + G^{liq}_{Ho} - 214080 + 44.108T$	This work
	${}^0G^{Liquid}_{Ho,HoTe} = -5352$	This work
	${}^1G^{Liquid}_{Ho,HoTe} = +1972$	This work
	${}^0G^{Liquid}_{HoTe,Te} = -90153 + 47.255T$	This work
HoTe		
	${}^0G^{HoTe}_{Ho:Te} = +0.5G^{SER}_{Ho} + 0.5G^{SER}_{Te} - 150323 + 41.574T$	This work
	${}^0G^{HoTe}_{Va:Te} = +0.5G^{SER}_{Te} + 224951$	This work
	${}^0G^{HoTe}_{Ho,Va:Te} = -363801 + 63.916T$	This work
Ho_2Te_5		
	${}^0G^{Ho_2Te_5}_{Ho:Te} = +0.2857G^{SER}_{Ho} + 0.7143G^{SER}_{Te} - 107479 + 36.622T$	This work
Ho_2Te_5		
	${}^0G^{HoTe_3}_{Ho:Te} = +0.25G^{SER}_{Ho} + 0.75G^{SER}_{Te} - 94022 + 30.919T$	This work

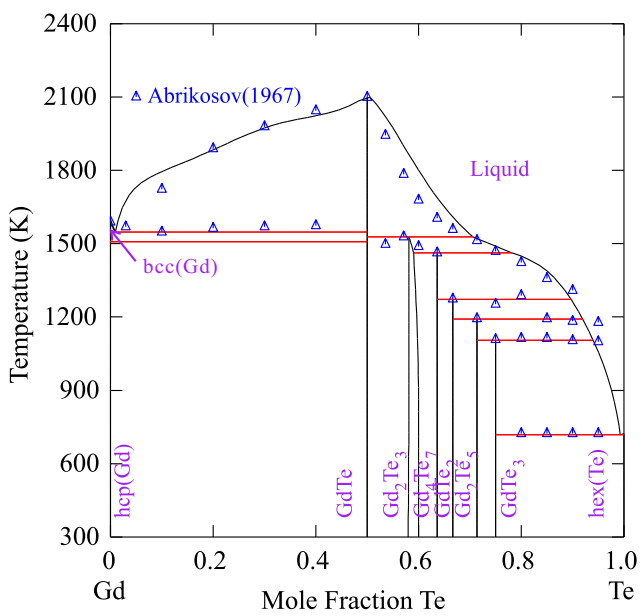

Fig. 1 The calculated phase diagram of the Gd-Te binary system with the experimental data

The calculated formation enthalpies of the compounds in the Gd-Te binary systems are shown in Fig. 2. As can be seen, the formation enthalpies calculated using CALPHAD method in this work are close to the experimental ones.[32] It is worth pointing out that all the thermodynamic data are given the equal weight in our calculation. The calculated formation entropies of the compounds in the Gd-Te binary systems are presented in Fig. 3. It is clear that a reasonable agreement is obtained between the calculated and experimental data.[32]

The mixing enthalpies which are calculated using CALPHAD and AIMD simulations in this work are presented in Fig. 4. The difference between the mixing enthalpies calculated in different methods is not significant. The difference of the calculated results between the CALPHAD method and the AIMD simulations becomes larger near 50 at.% Te, which can be attributed to the small temperature difference between the melting point of GdTe (2098 K) and the AIMD simulations (2280 K). It is hard to reach equilibrium within 48 picoseconds (16000 steps) in AIMD simulation.

4.2 The Dy-Te Binary System

The calculated Dy-Te binary phase diagram in comparison with the experimental data[27] is shown in Fig. 5 and Table 3. It is obvious that the presently obtained thermodynamic parameters can reproduce the experimental data well. According to our calculation, the slope of the liquidus flattens around the composition of 10 at.% Te and the temperature of liquidus is higher than the experimental one. In addition, an attempt was made to give the

Table 3 Experimental and calculated invariant reactions in the Gd-Te, Dy-Te and Ho-Te systems

Invariant Reactions	Temperature, K	Composition, at.% Te			Refs.
Gd-Te binary system					
Liquid ↔ bcc(Gd) + GdTe	1563	2	0	50	[31]
	1548	1.0	0	50	This work
Liquid ↔ GdTe	2098	50	…	…	[31]
	2097	50	…	…	This work
Liquid + GdTe ↔ Gd_2Te_3	1528	70	50	57	[31]
	1528	70.4	50	58.1	This work
Liquid + Gd_2Te_3 ↔ Gd_4Te_7	1463	73	60	64	[31]
	1462	78.2	59.0	64	This work
Liquid + Gd_4Te_7 ↔ $GdTe_2$	1273	90	63.6	66.7	[31]
	1272	89.7	63.6	66.7	This work
Liquid + $GdTe_2$ ↔ Gd_2Te_5	1193	91	66.7	71.4	[31]
	1190	91.7	66.7	71.4	This work
Liquid + Gd_2Te_5 ↔ $GdTe_3$	1108	92	71.4	75	[31]
	1104	94.0	71.4	75	This work
Liquid ↔ $GdTe_3$ + hex(Te)	723	…	75	100	[31]
	719	~ 99	75	100	This work
Dy-Te binary system					
Liquid ↔ bcc(Dy) + DyTe	1658	3	0	50	[34]
	1657	0.7	0	50	This work
Liquid ↔ DyTe	2123	50	…	…	(Ref34]
	2113	50	…	…	This work
Dy_2Te ↔ hcp(Dy) + DyTe	1343	33	0	50	[26]
	1342	33	0	50	This work
Liquid ↔ DyTe + Dy_2Te_3	1813	56	50	60	[34]
	1820	56	50	60	This work
Liquid ↔ Dy_2Te_3	1843	60	…	…	[34]
	1838	60	…	…	This work
Liquid + Dy_2Te_3 ↔ Dy_4Te_7	1503	65	60	63.6	[34]
	1496	66.5	59.2	63.6	This work
Liquid + Dy_4Te_7 ↔ Dy_4Te_9	1193	87	63.6	69.2	[34]
	1192	86.7	63.6	69.2	This work
Liquid + Dy_4Te_9 ↔ Dy_4Te_{11}	1128	90	69.2	73.3	[34]
	1128	90.7	69.2	73.3	This work
Liquid ↔ Dy_4Te_{11} + hex(Te)	728	…	73.3	100	[34]
	723	100	73.3	100	This work
Ho-Te binary system					
Liquid ↔ hcp(Ho) + HoTe	1383	23	…	48.5	[29]
	1383	23	…	50	[35]
	1387	20.4	…	50	This work
Liquid ↔ HoTe	1643	…	50	…	[29]
	1642	…	50	…	[35]
	1642	…	50.8	…	This work
Liquid + HoTe ↔ Dy_2Te_5	1073	…	54.7	71.4	[29]
	1073	93	54.7	71.4	[35]
	1073	91.8	53.6	71.4	This work
Liquid + Ho_2Te_5 ↔ $HoTe_3$	1053	…	71.4	75	[29]
	1053	94.1	71.4	75	[35]

Table 3 continued

Invariant Reactions	Temperature, K	Composition, at.% Te			Refs.
	1052	93.4	71.4	75	This work
Liquid ↔ $HoTe_3$ + hex(Te)	728	…	…	75	[29]
	722	~ 99.9	100	75	[35]
	722	~ 99.9	100	75	This work

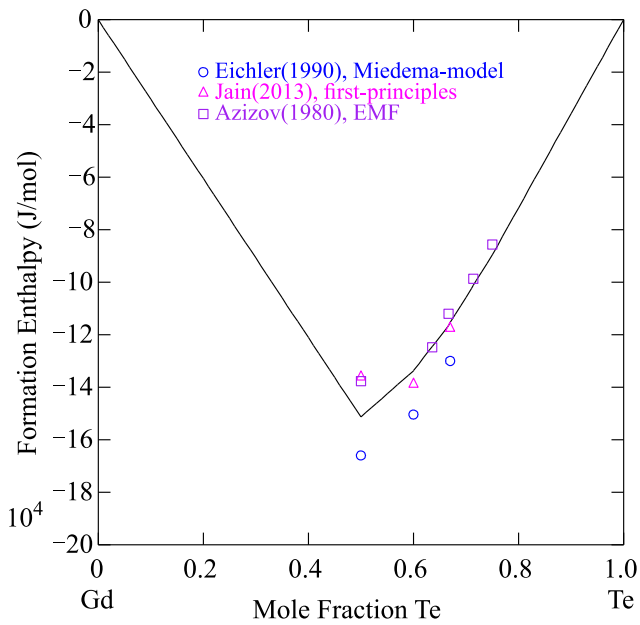

Fig. 2 The calculated formation enthalpies of compounds in the Gd-Te binary system at 298.15 K with the experimental data from Eicvhler et. al.,[34] Jain et al.[33] and Azizov et al.[32] The reference state is hcp for Gd and hex for Te

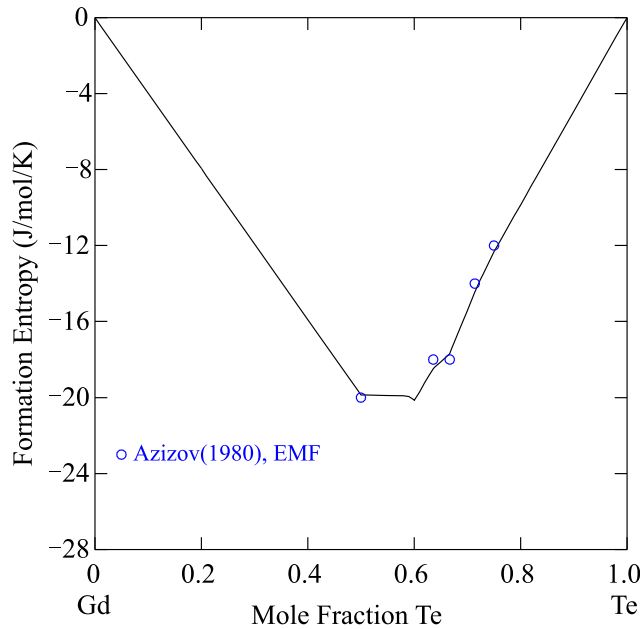

Fig. 3 The calculated formation entropies of compounds in the Gd-Te binary system at 298.15 K with the experimental data from Azizov and Zargarian.[32] The reference state is hcp for Gd and hex for Te

experimental data (liquidus temperature at 10 at %.Te) a higher weight in our assessment. However, a liquid-liquid separation appears. Therefore, it is predicted that liquid-liquid separation may exist around 10%.Te in the Dy-Te system in the present work. The compound Dy_2Te is considered in this work. According to the experimental data, Dy_2Te decomposes into DyTe and hcp(Dy) above about 1343 K, which is also reflected by our calculated results.

The calculated formation enthalpies are shown in Fig. 6, which agree well with the ones reported in the literature.[19] Figure 7 presents the calculated mixing enthalpies using the CALPHAD method and the AIMD simulations in this work. In our assessment, all the mixing enthalpies data are given the same weight. It can be seen that most CALPHAD results are close to the AIMD ones except for the composition of x(Te) = 0.1. It was also attempted to give mixing enthalpy data a heigh weight and it also causes a liquid-liquid separation, which also indicates the existence of the liquid-liquid separation in the Dy-Te system.

4.3 The Ho-Te Binary System

As shown in Fig. 8, the Ho-Te calculated phase diagram is compared with the experimental data[29] and the calculation results reported by Ghamri et al.[35] The calculated phase diagram agrees well with the experimental data.[29] In our reassessment, the parameters reported by Ghamri et al.[35] are set as start values. It is worth to mention that the regular solution model was initially adopted to describe the liquid phase and then the associate model is used to describe the

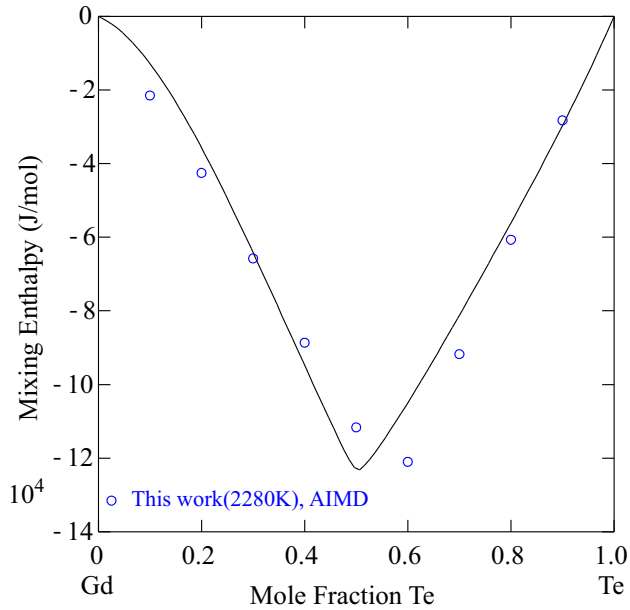

Fig. 4 The calculated mixing enthalpies of liquid phase in the Gd-Te binary system at 2280 K. The reference state is liquid for Gd and Te

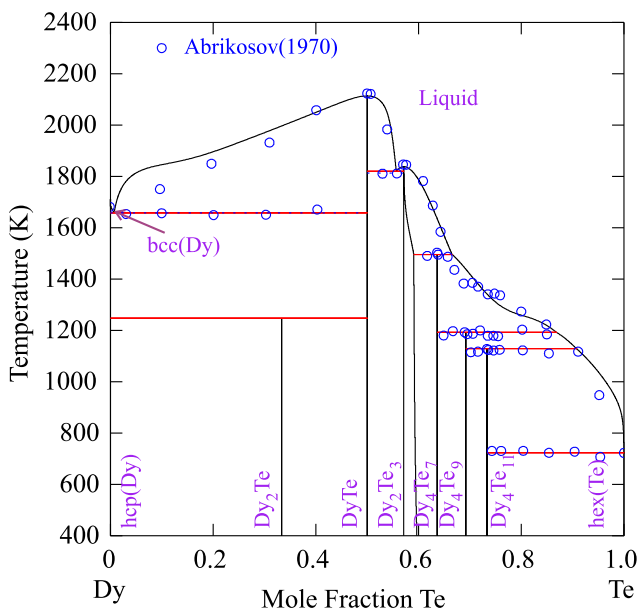

Fig. 5 The calculated phase diagram of the Dy-Te binary system with the experimental data

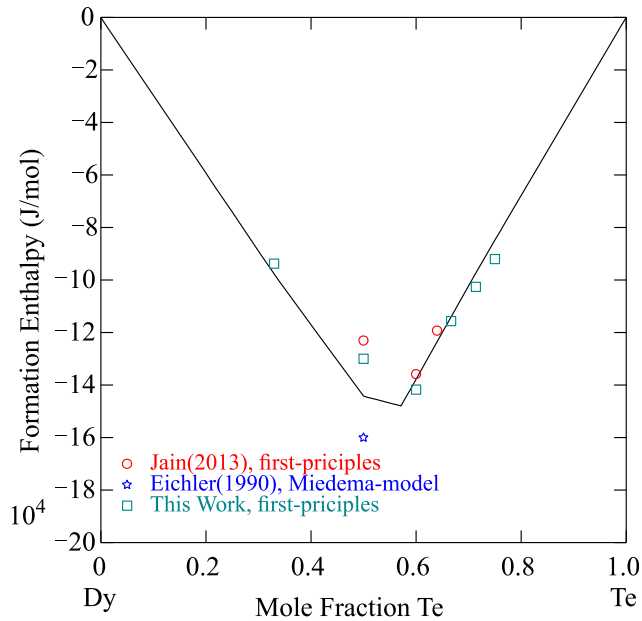

Fig. 6 The calculated formation enthalpies of compounds in the Dy-Te binary system at 298.15 K with the experimental data from Jain et al.[33] and Eichler et al.[34] The reference state is hcp for Dy and hex for Te

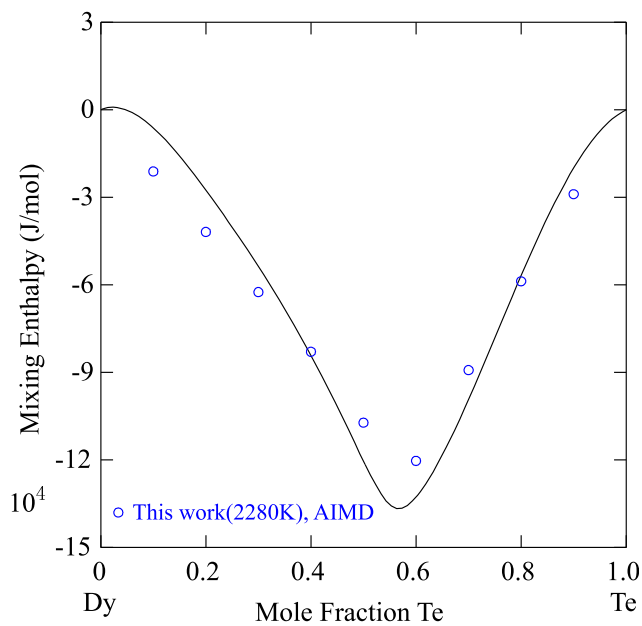

Fig. 7 The calculated mixing enthalpies of liquid phase in the Dy-Te binary system at 2280 K. The reference state is liquid for Dy and Te

liquid phase in order to reproduce the low mixing enthalpies calculated in this work. A steeply rising liquidus is obtained on the Ho-rich side.

The calculated formation enthalpies and mixing enthalpies are shown in Fig. 9 and 10. The calculated formation enthalpies are close to the results calculated by the first-principles method.[35] The mixing enthalpies calculated in this work using the CALPHAD method agree well with the ones calculated using the AIMD simulations, which are lower than the results performed by Ghamri et al.[35]

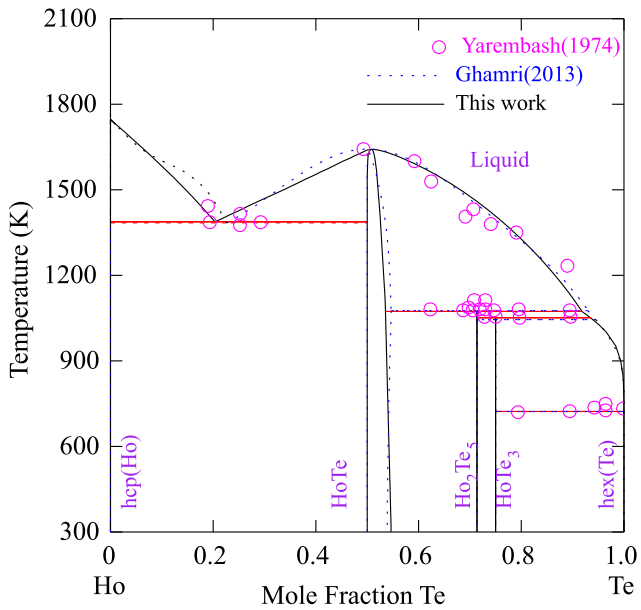

Fig. 8 The calculated phase diagram of the Ho-Te binary system compared with the experimental data from Yarembash et al.[29] and the calculated one from Ghamri et al.[35]

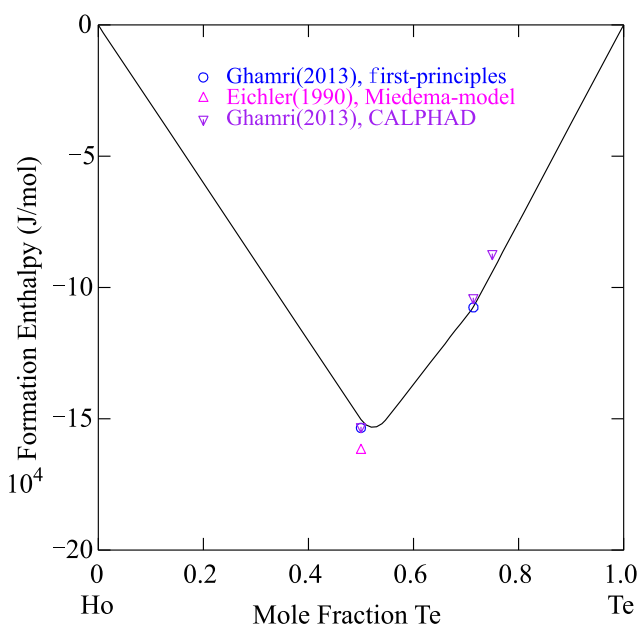

Fig. 9 The calculated formation enthalpies of compounds in the Ho-Te binary system at 298.15 K with the experimental data from Ghamri et. al.[35] and Eichler et al.[34] The reference state is hcp for Ho and hex for Te

5 Conclusion

Based on the phase equilibrium and thermodynamic data reported in the literature, the calculated mixing enthalpies of the liquid phases and the formation enthalpy of the compounds, the Gd-Te, Dy-Te and Ho-Te binary systems

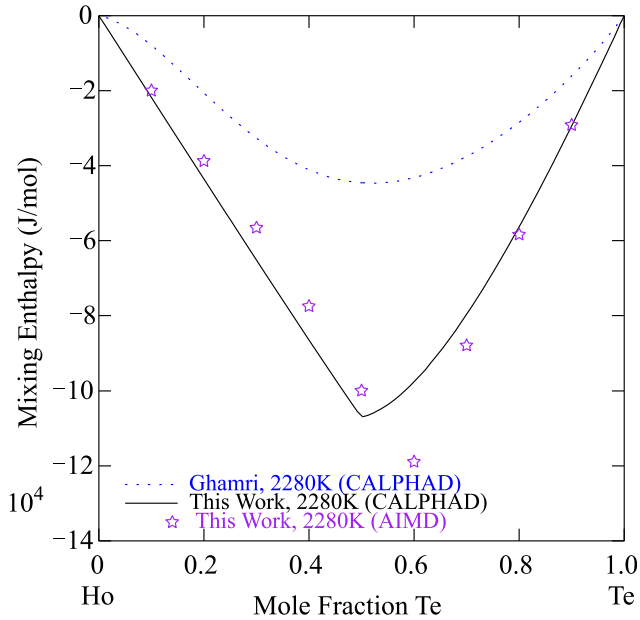

Fig. 10 The calculated mixing enthalpies of liquid phase in the Ho-Te binary system at 2280 K compared with the results calculated by Ghamri et al.[35] The reference state is liquid for Ho and Te

are assessed thermodynamically. Finally, a set of consistent thermodynamic parameters for the Gd-Te, Dy-Te and Ho-Te systems are established. The calculated phase diagrams and thermodynamic properties agree well with the ones reported in the literature and those calculated in this work. In addition, it is predicted that there exists a liquid-liquid separation in the Dy-rich side in the Dy-Te binary system.

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